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3	Write a program to fill all missing values in a DataFrame with zero.										
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5	Write a program to draw bar graph for the following data for the Medal tally of CWG-2018:- <table border="1" data-bbox="256 804 1038 913"> <tr> <td>Gold</td><td>Silver</td><td>Bronze</td><td>Total</td></tr> <tr> <td>26</td><td>20</td><td>20</td><td>66</td></tr> </table>	Gold	Silver	Bronze	Total	26	20	20	66		
Gold	Silver	Bronze	Total								
26	20	20	66								
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Program 1

Write a program to create a DataFrame have E-commerce data and perform selection of row/column using loc() and iloc().

In [2]: `import pandas as pd`

```
data = {
    'Order_ID': [101, 102, 103, 104, 105],
    'Product': ['Laptop', 'Mobile', 'Headphones', 'Tablet', 'Camera'],
    'Price': [50000, 20000, 2000, 15000, 30000],
    'Quantity': [1, 2, 3, 1, 2],
    'Customer': ['Amit', 'Riya', 'John', 'Sara', 'Rahul']
}

df = pd.DataFrame(data)
print(df)
```

	Order_ID	Product	Price	Quantity	Customer
0	101	Laptop	50000	1	Amit
1	102	Mobile	20000	2	Riya
2	103	Headphones	2000	3	John
3	104	Tablet	15000	1	Sara
4	105	Camera	30000	2	Rahul

In [3]: `print(df.loc[2])`
`print(df.loc[:, ['Product', 'Price']])`

```
Order_ID      103
Product      Headphones
Price          2000
Quantity          3
Customer      John
Name: 2, dtype: object
   Product  Price
0   Laptop  50000
1   Mobile  20000
2 Headphones   2000
3   Tablet  15000
4   Camera  30000
```

In [4]: `print(df.iloc[0])`
`print(df.iloc[1:4, 1:4])`

```
Order_ID      101
Product      Laptop
Price          50000
Quantity          1
Customer      Amit
Name: 0, dtype: object
   Product  Price  Quantity
1   Mobile  20000          2
2 Headphones   2000          3
3   Tablet  15000          1
```

Program 2

Create a Series object S5 containing numbers. Write a program to store the square of the series values in object S6. Display S6's values which are >15.

```
In [1]: import pandas as pd

S5 = pd.Series([1, 2, 3, 4, 5, 6])

S6 = S5 ** 2

print(S6[S6 > 15])
```

```
3    16
4    25
5    36
dtype: int64
```

Program 3

Write a program to fill all missing values in a DataFrame with zero.

```
In [1]: import pandas as pd
import numpy as np

data = {
    'A': [1, np.nan, 3],
    'B': [np.nan, 5, 6],
    'C': [7, 8, np.nan]
}

df = pd.DataFrame(data)

df = df.fillna(0)

print(df)
```

	A	B	C
0	1.0	0.0	7.0
1	0.0	5.0	8.0
2	3.0	6.0	0.0

```
In [ ]:
```

Program 4

Program for combining DataFrames using concat(), join(), merge().

```
In [1]: import pandas as pd

df1 = pd.DataFrame({
    'ID': [1, 2, 3],
    'Name': ['Amit', 'Riya', 'John']
})

df2 = pd.DataFrame({
    'ID': [1, 2, 4],
    'Marks': [85, 90, 75]
})

concat_df = pd.concat([df1, df2], axis=0)

join_df = df1.join(df2.set_index('ID'), on='ID')

merge_df = pd.merge(df1, df2, on='ID')

print(concat_df)
print(join_df)
print(merge_df)
```

```
   ID  Name  Marks
0   1  Amit   NaN
1   2  Riya   NaN
2   3  John   NaN
0   1   NaN  85.0
1   2   NaN  90.0
2   4   NaN  75.0
   ID  Name  Marks
0   1  Amit  85.0
1   2  Riya  90.0
2   3  John   NaN
   ID  Name  Marks
0   1  Amit    85
1   2  Riya    90
```

```
In [ ]:
```

Program 5

Write a program to draw bar graph for the following data for the Medal tally of CWG-2018.

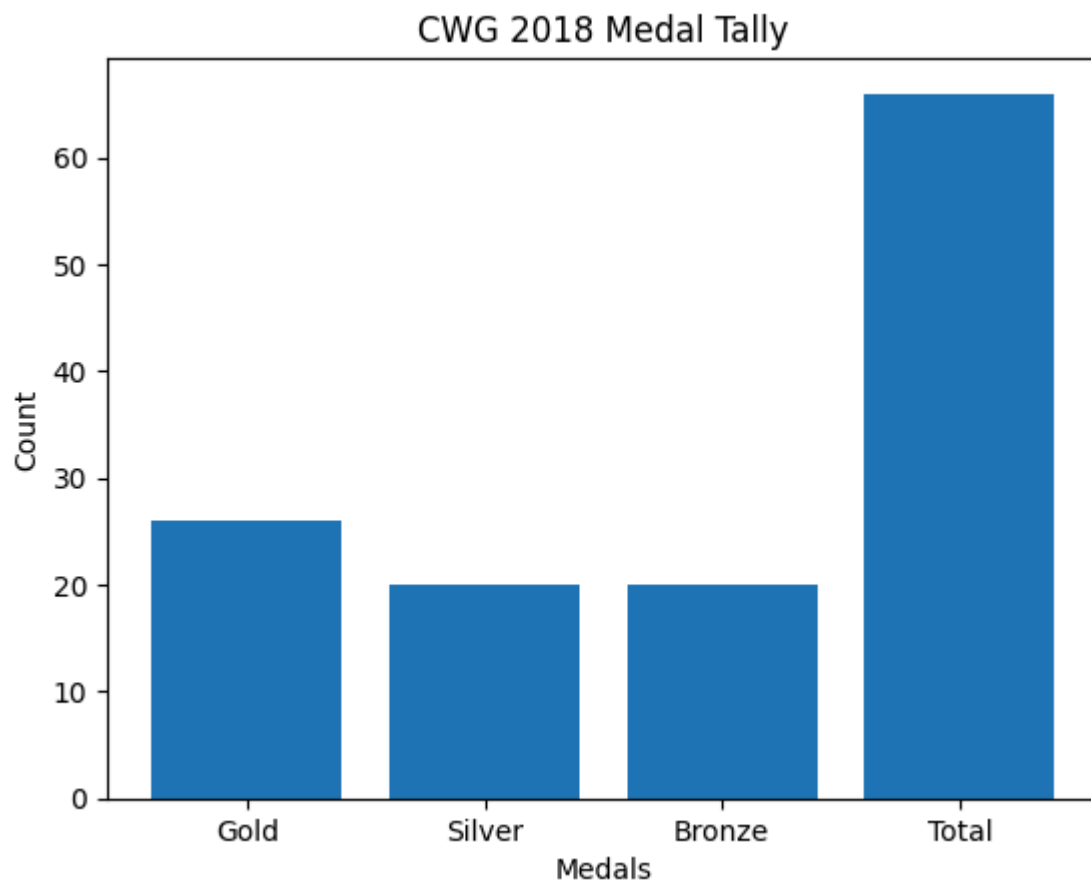
```
In [1]: import matplotlib.pyplot as plt

medals = ['Gold', 'Silver', 'Bronze', 'Total']
values = [26, 20, 20, 66]

plt.bar(medals, values)

plt.xlabel('Medals')
plt.ylabel('Count')
plt.title('CWG 2018 Medal Tally')

plt.show()
```



```
In [ ]:
```

Program 6

Implementing Line plot, Dist plot, Lmplot, Count plot using Seaborn library.

```
In [1]: import seaborn as sns
import matplotlib.pyplot as plt

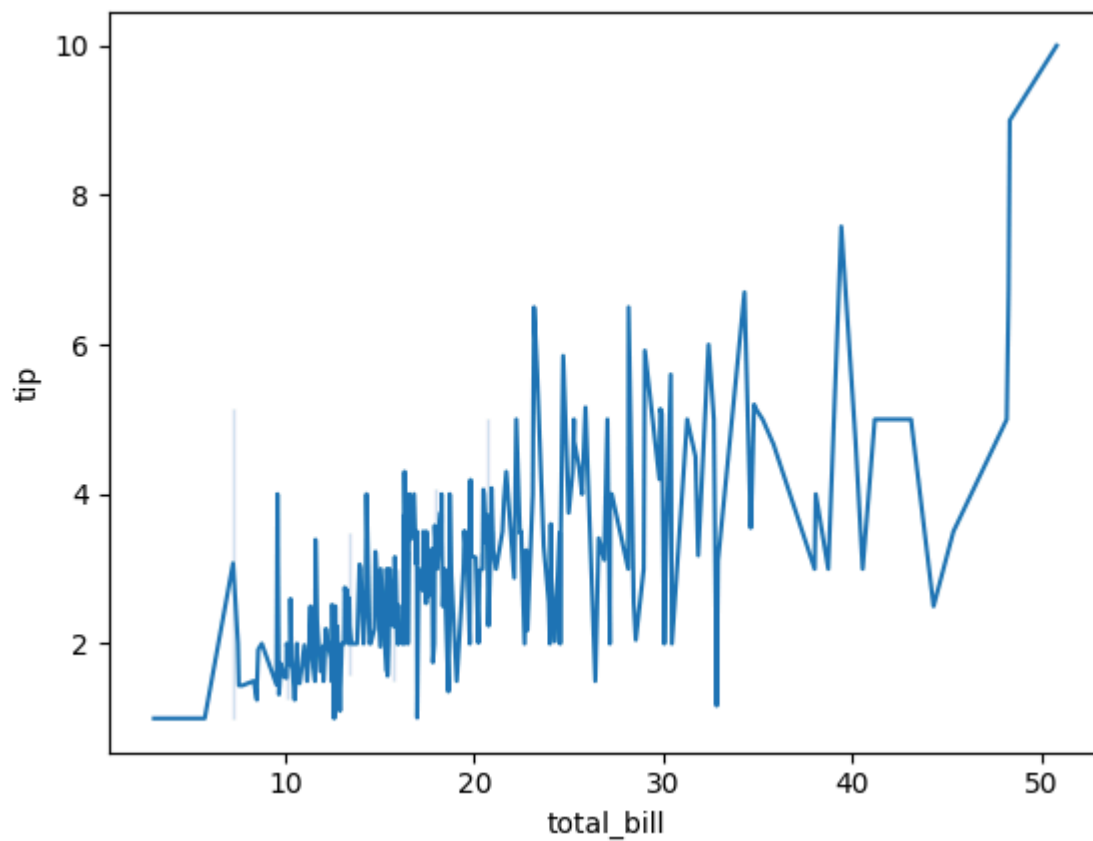
data = sns.load_dataset("tips")

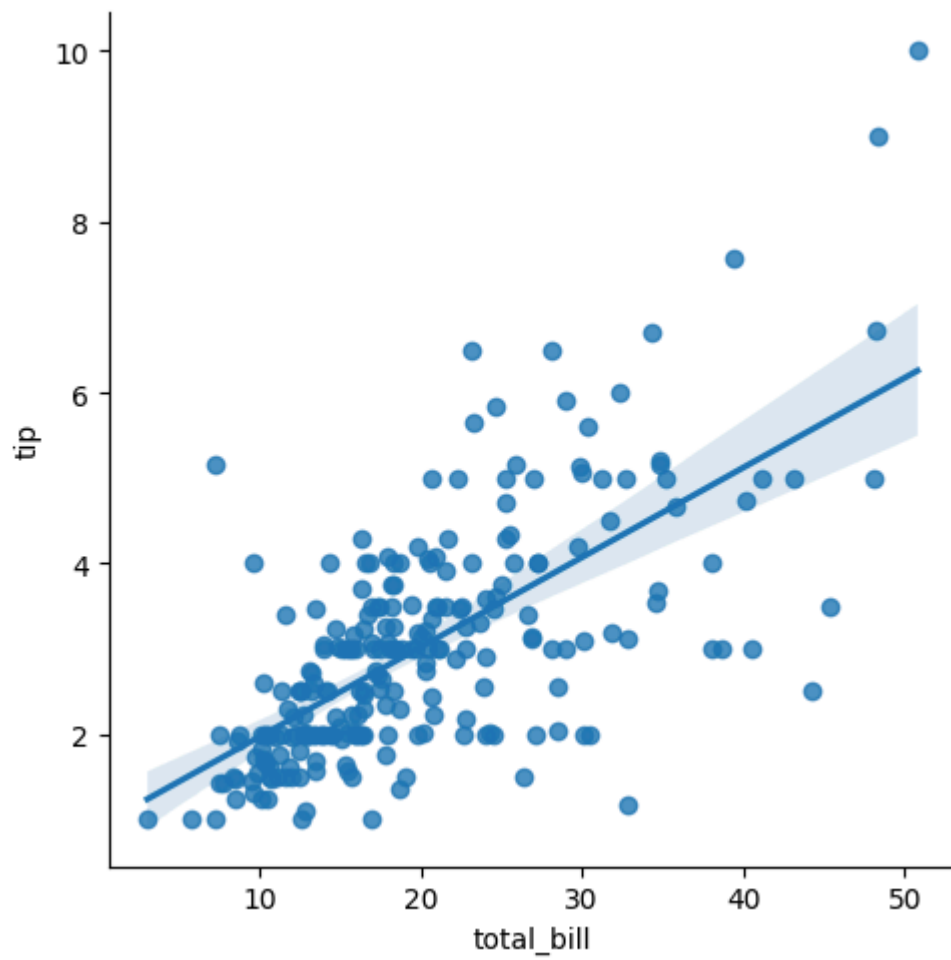
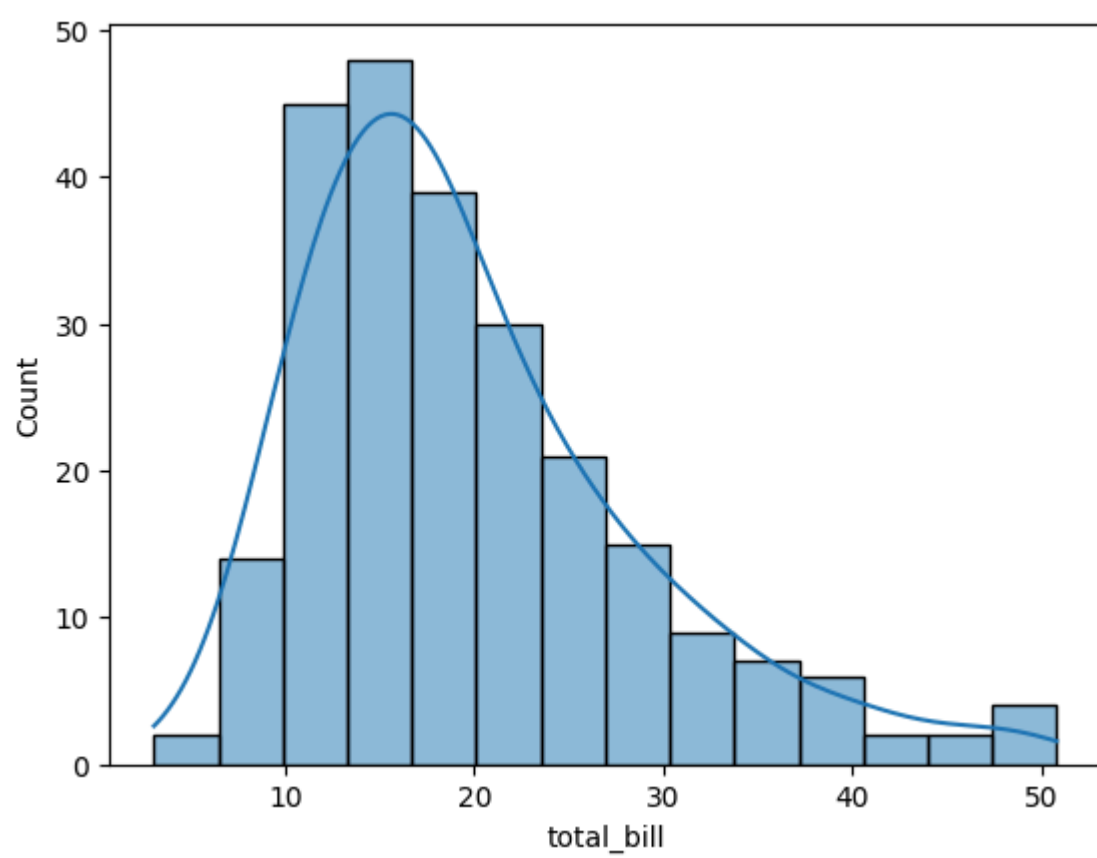
sns.lineplot(x="total_bill", y="tip", data=data)
plt.show()

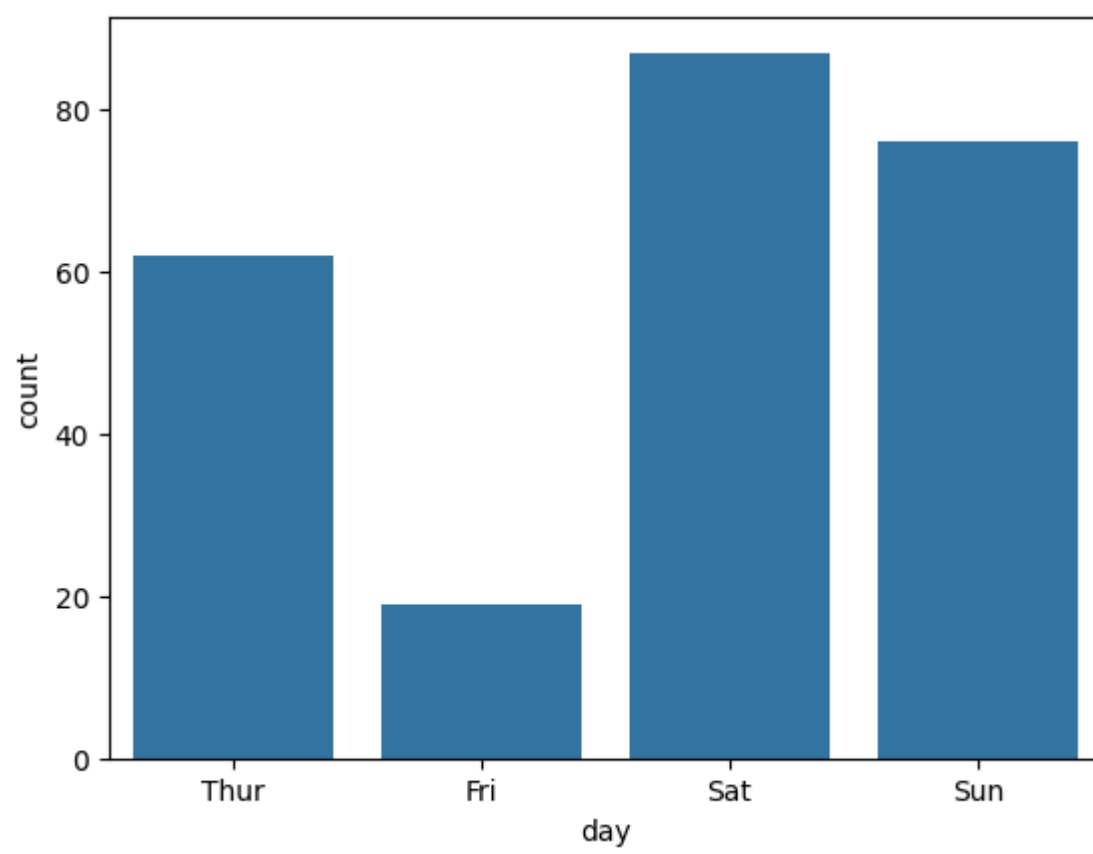
sns.histplot(data["total_bill"], kde=True)
plt.show()

sns.lmplot(x="total_bill", y="tip", data=data)
plt.show()

sns.countplot(x="day", data=data)
plt.show()
```







In []:

Program 7

Create a DataFrame namely *aid* that stores aid (Toys, books, uniform, shoes) by NGO's for different states. Write a program to display the aid for:- a) Books & Uniform Only
b) Shoes Only

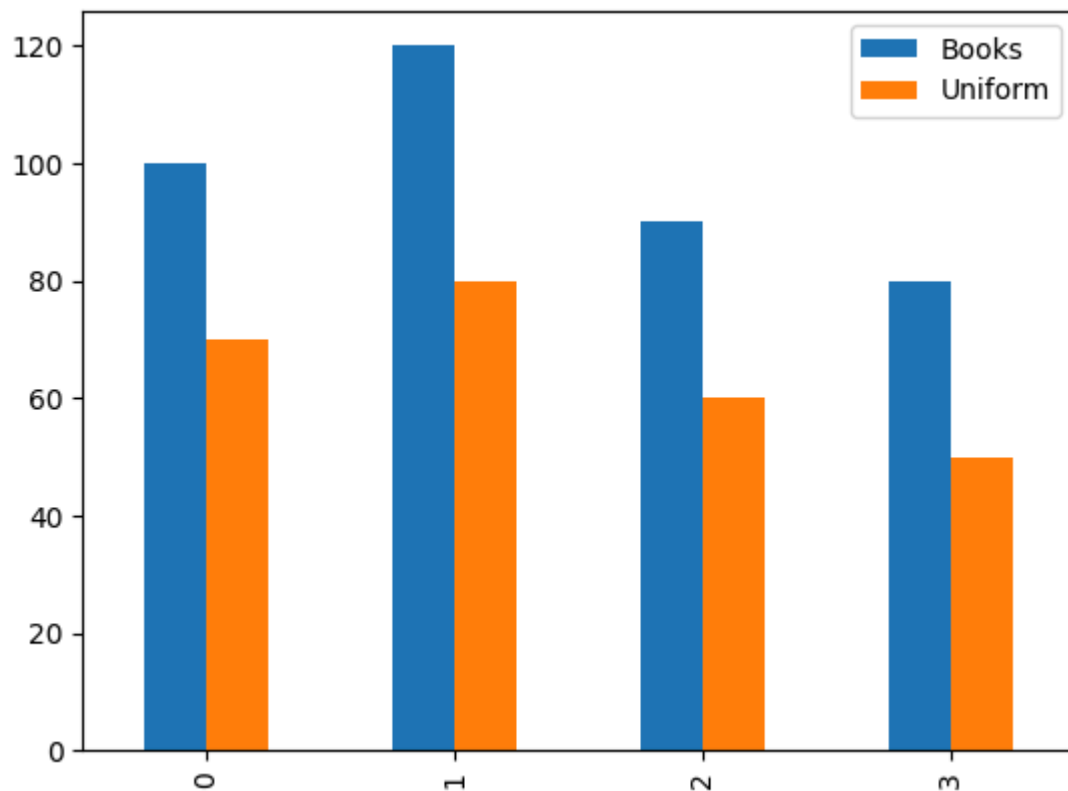
```
In [2]: import pandas as pd
import matplotlib.pyplot as plt

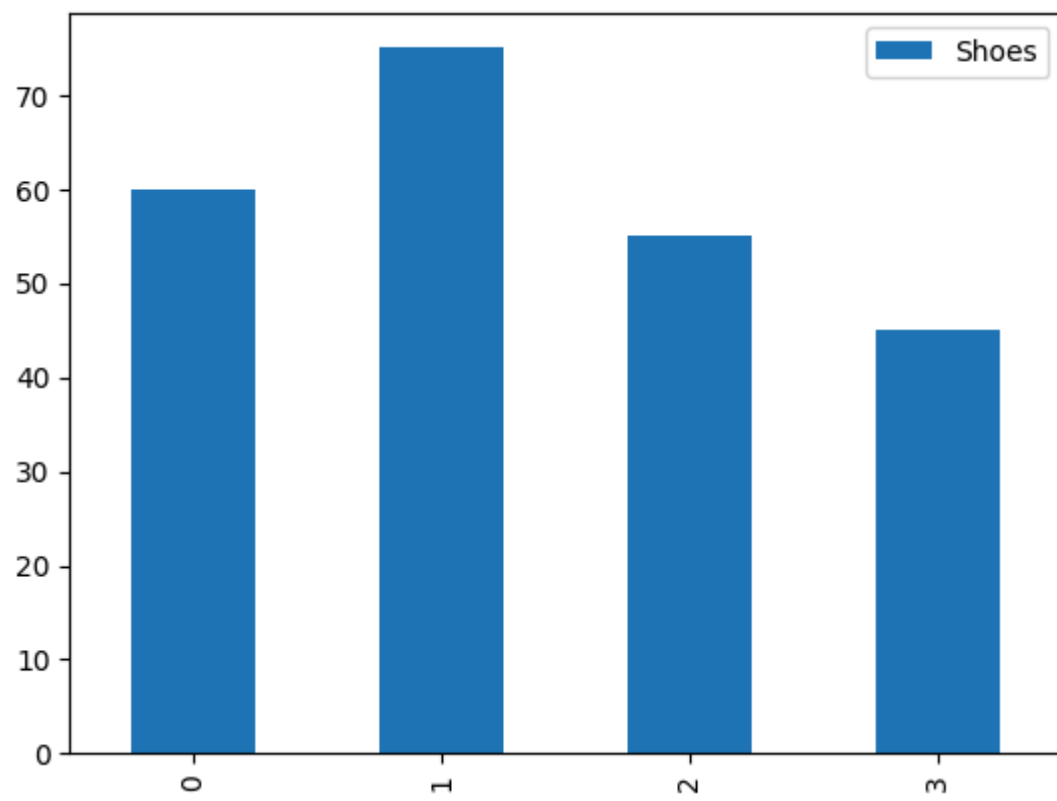
data = {
    'State': ['Delhi', 'Maharashtra', 'Punjab', 'Gujarat'],
    'Toys': [50, 60, 40, 30],
    'Books': [100, 120, 90, 80],
    'Uniform': [70, 80, 60, 50],
    'Shoes': [60, 75, 55, 45]
}

aid = pd.DataFrame(data)

aid[['Books', 'Uniform']].plot(kind='bar')
plt.show()

aid[['Shoes']].plot(kind='bar')
plt.show()
```





In []:

Program 8

Create a DataFrame *ndf* having Name, Gender, Position, City, Age, Projects. Write a program to summarize how many projects are being handled by each position for each city?

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt

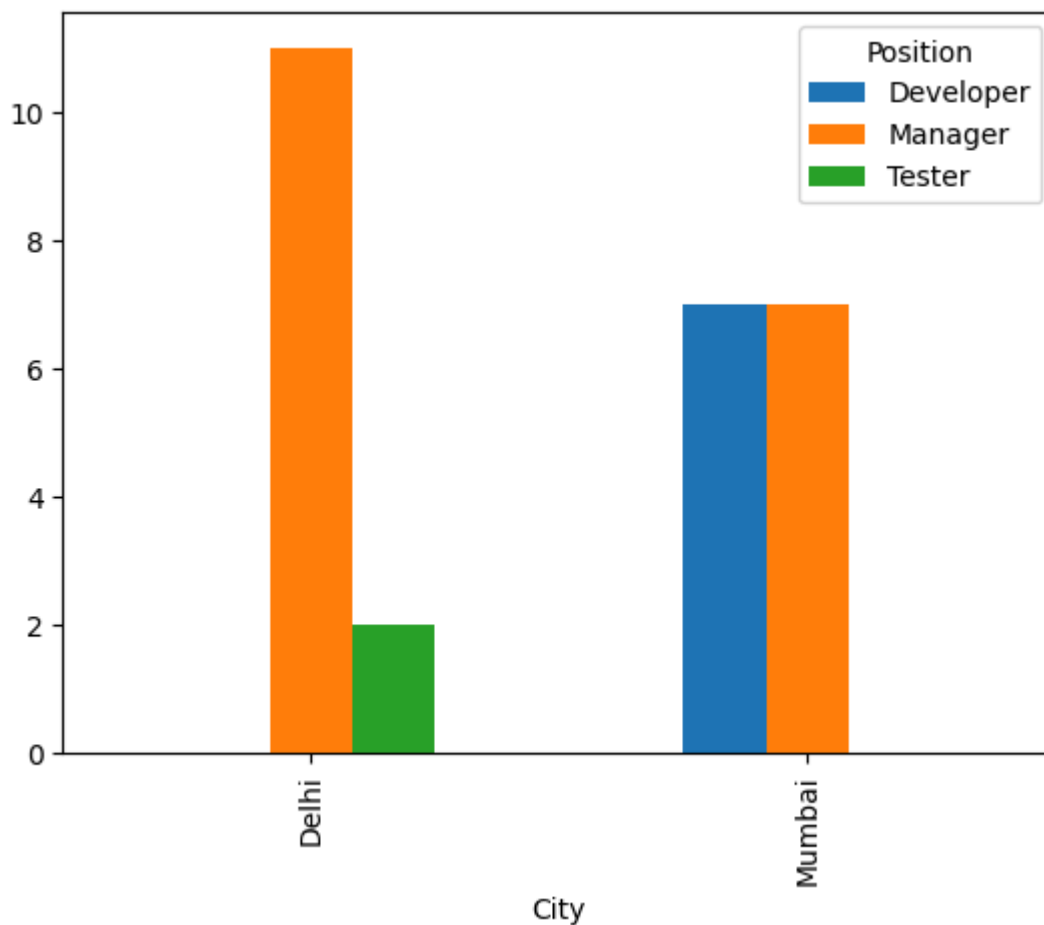
data = {
    'Name': ['Amit', 'Riya', 'John', 'Sara', 'Rahul', 'Neha'],
    'Gender': ['M', 'F', 'M', 'F', 'M', 'F'],
    'Position': ['Manager', 'Developer', 'Manager', 'Developer', 'Tester', 'Manager'],
    'City': ['Delhi', 'Mumbai', 'Delhi', 'Mumbai', 'Delhi', 'Mumbai'],
    'Age': [30, 28, 35, 26, 27, 32],
    'Projects': [5, 3, 6, 4, 2, 7]
}

ndf = pd.DataFrame(data)

result = ndf.groupby(['City', 'Position'])['Projects'].sum().unstack()

result.plot(kind='bar')

plt.show()
```



In []:

Program 9

Marks is a list that stores marks of a student in 10 unit test. Write a program to plot Line chart for the student's performance in these 10 test

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt

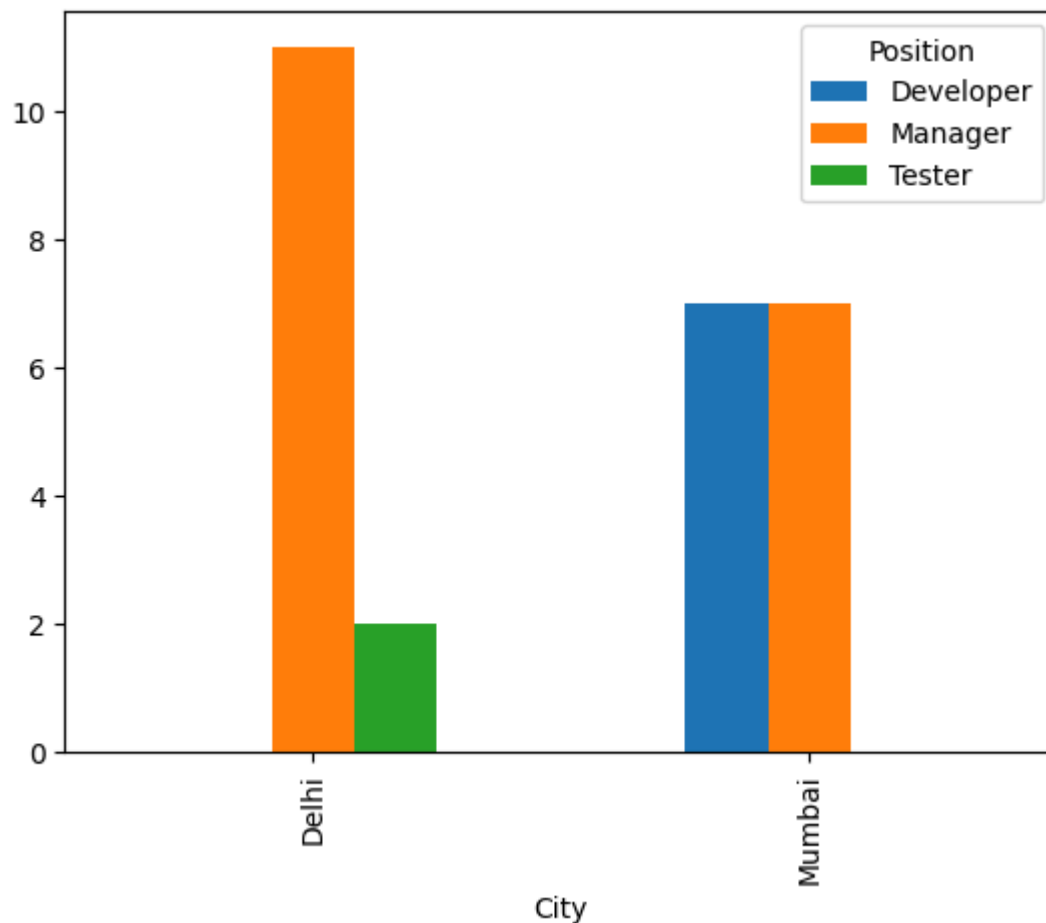
data = {
    'Name': ['Amit', 'Riya', 'John', 'Sara', 'Rahul', 'Neha'],
    'Gender': ['M', 'F', 'M', 'F', 'M', 'F'],
    'Position': ['Manager', 'Developer', 'Manager', 'Developer', 'Tester', 'Manager'],
    'City': ['Delhi', 'Mumbai', 'Delhi', 'Mumbai', 'Delhi', 'Mumbai'],
    'Age': [30, 28, 35, 26, 27, 32],
    'Projects': [5, 3, 6, 4, 2, 7]
}

ndf = pd.DataFrame(data)

result = ndf.groupby(['City', 'Position'])['Projects'].sum().unstack()

result.plot(kind='bar')

plt.show()
```



In []:

Program 10

Write a program to plot a horizontal bar chart from the height of some students

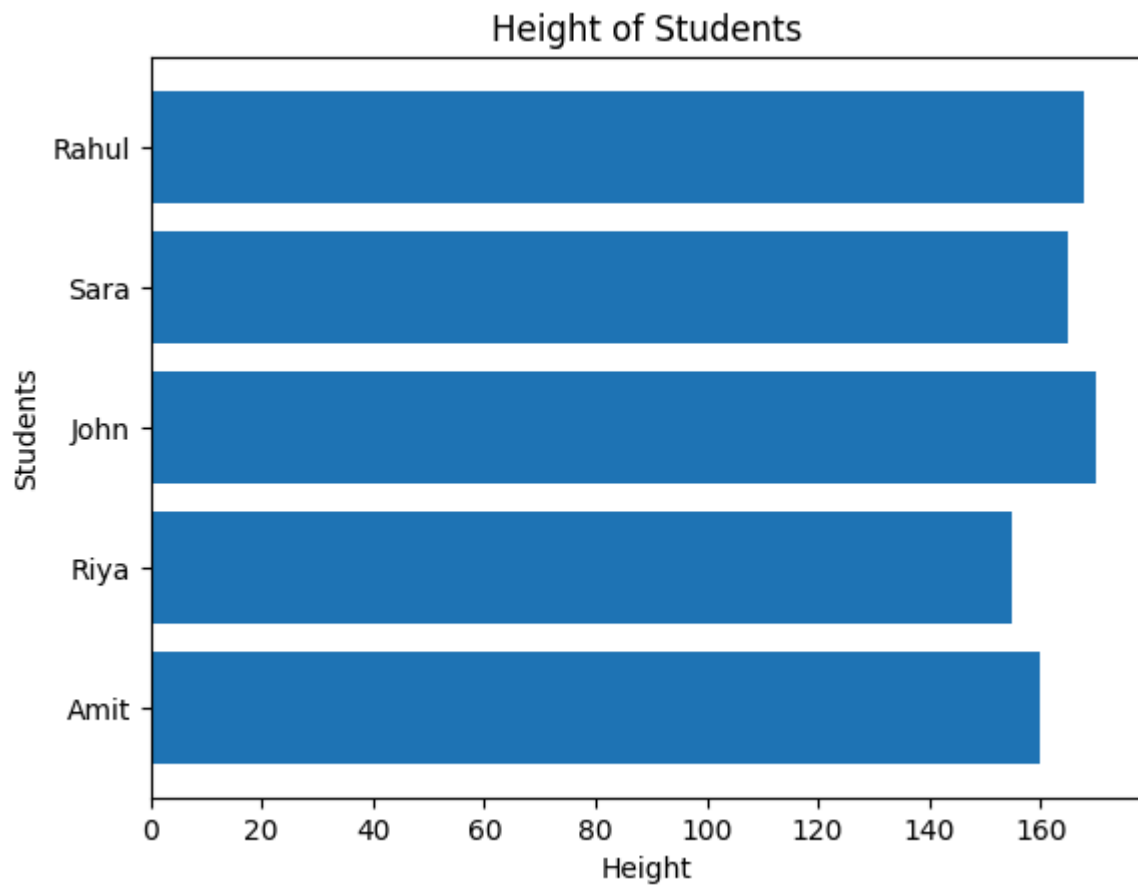
```
In [1]: import matplotlib.pyplot as plt

students = ['Amit', 'Riya', 'John', 'Sara', 'Rahul']
heights = [160, 155, 170, 165, 168]

plt.barh(students, heights)

plt.xlabel('Height')
plt.ylabel('Students')
plt.title('Height of Students')

plt.show()
```



In []:

Program 11

Write a program to implement ANNOVA.

```
In [2]: import pandas as pd
from scipy import stats
import matplotlib.pyplot as plt

group1 = [23, 20, 22, 21, 24]
group2 = [30, 28, 27, 29, 31]
group3 = [35, 33, 34, 32, 36]

f_stat, p_value = stats.f_oneway(group1, group2, group3)

print(f_stat)
print(p_value)

data = [group1, group2, group3]

plt.boxplot(data, labels=['Group1', 'Group2', 'Group3'])

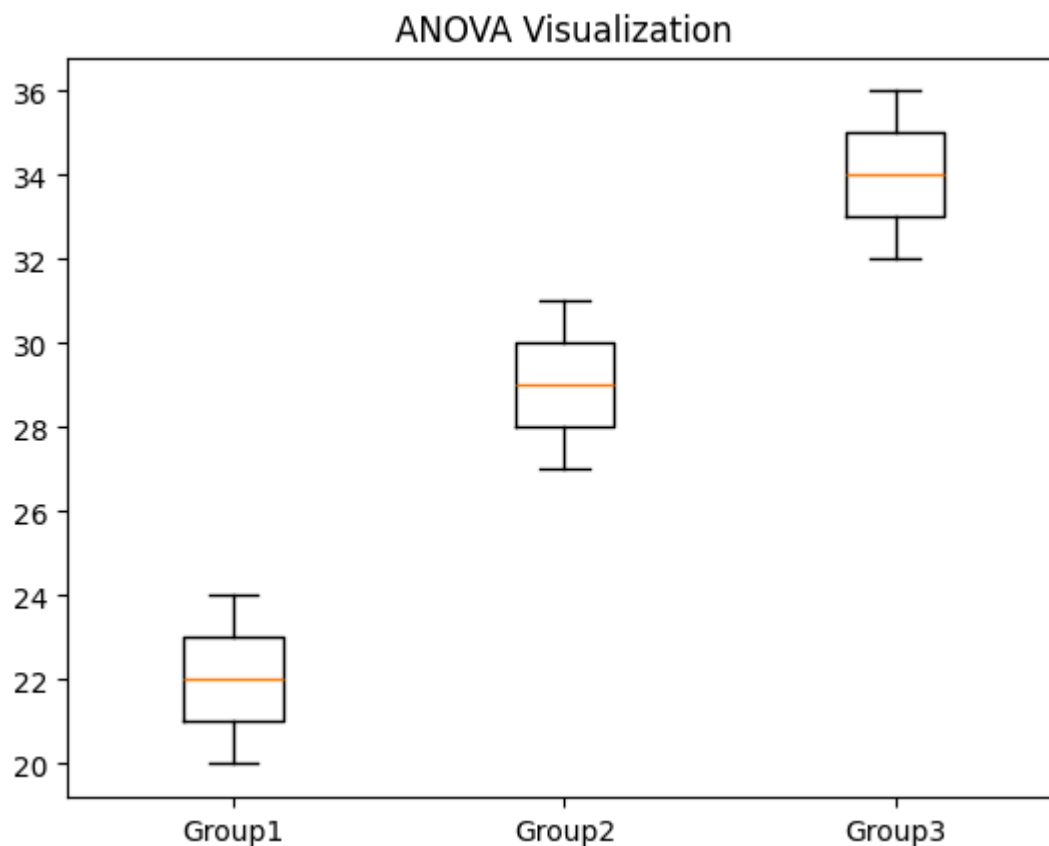
plt.title('ANOVA Visualization')
plt.show()
```

72.66666666666681

1.9686250761248166e-07

C:\Users\sujal\AppData\Local\Temp\ipykernel_12776\2263263624.py:16: MatplotlibDeprecationWarning: The 'labels' parameter of boxplot() has been renamed 'tick_labels' since Matplotlib 3.9; support for the old name will be dropped in 3.11.

```
plt.boxplot(data, labels=['Group1', 'Group2', 'Group3'])
```



In []:

Program 12

Write a program to show correlation between two randomly generated numbers.

```
In [1]: import numpy as np
import matplotlib.pyplot as plt

x = np.random.rand(50)
y = np.random.rand(50)

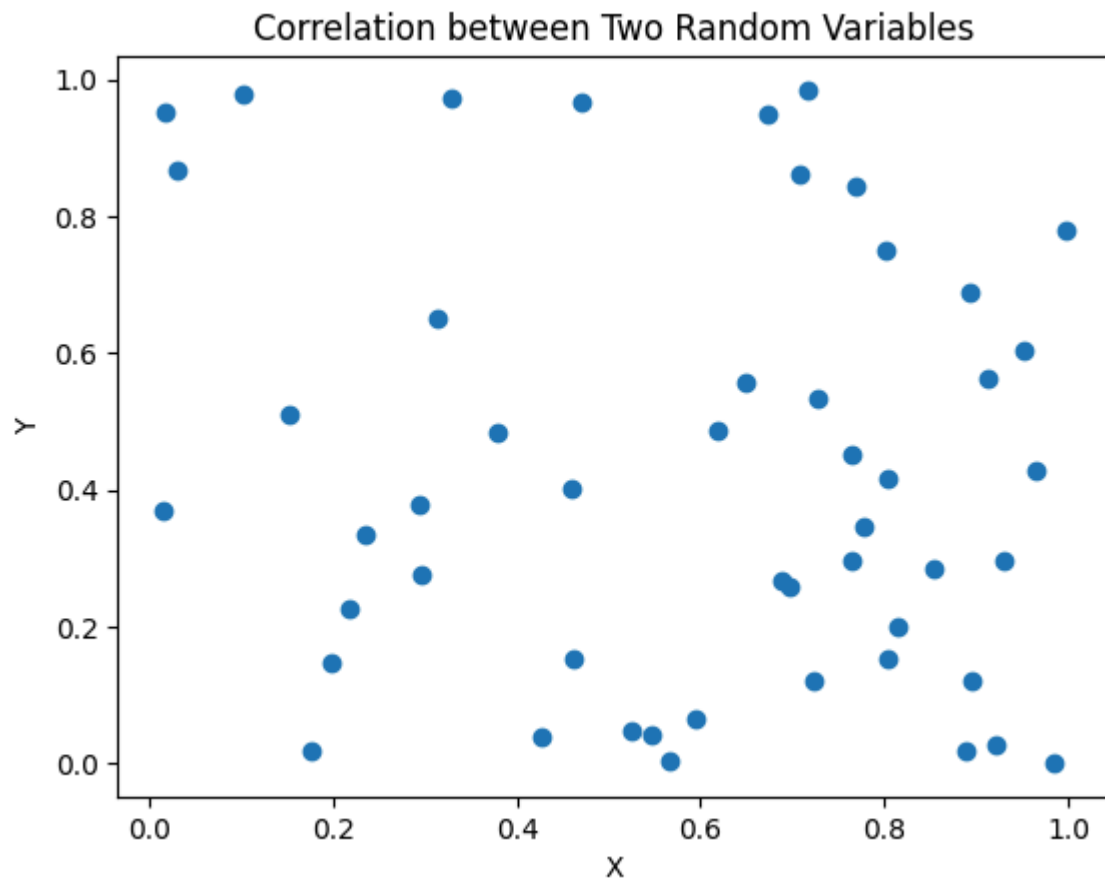
correlation = np.corrcoef(x, y)[0, 1]

print(correlation)

plt.scatter(x, y)
plt.title('Correlation between Two Random Variables')
plt.xlabel('X')
plt.ylabel('Y')

plt.show()
```

-0.1475970505716447



In []:

Program 13

Write a program to implement Covariance.

```
In [1]: import numpy as np
import matplotlib.pyplot as plt

x = np.array([10, 20, 30, 40, 50])
y = np.array([15, 25, 35, 45, 55])

covariance_matrix = np.cov(x, y)

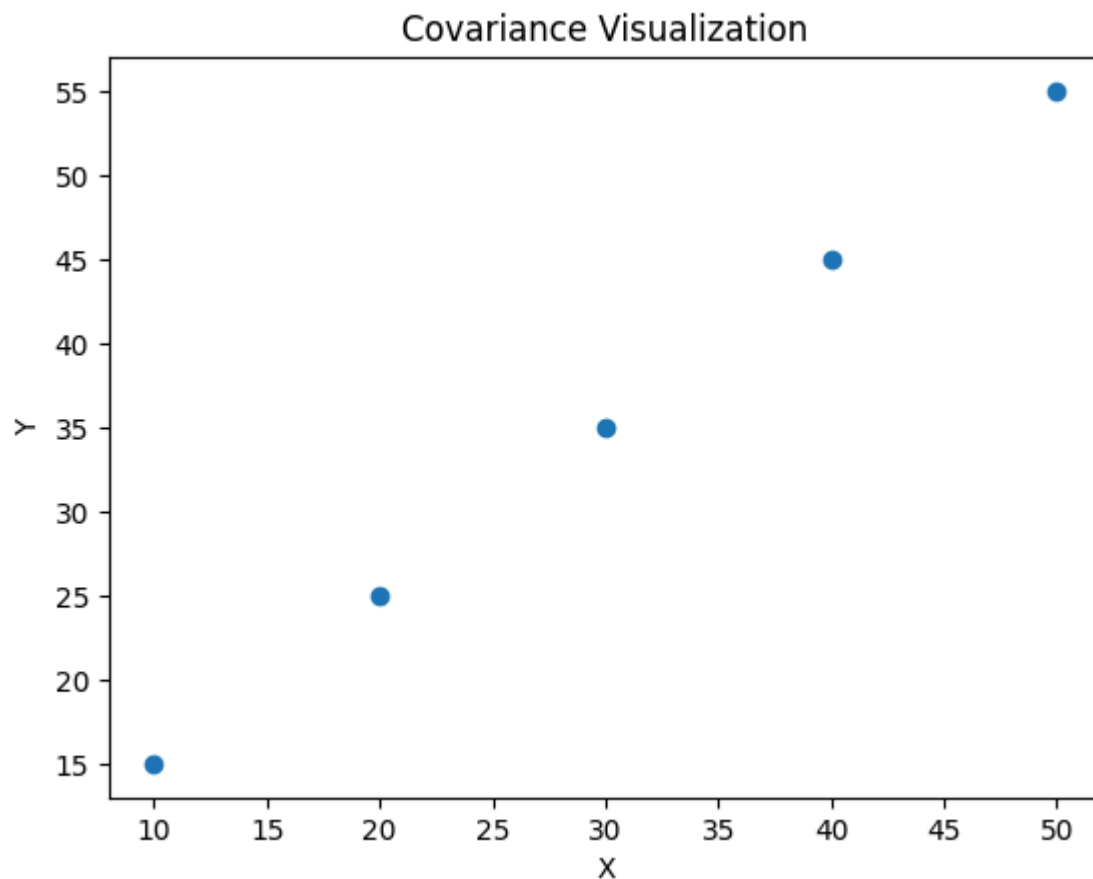
print(covariance_matrix)

plt.scatter(x, y)

plt.title('Covariance Visualization')
plt.xlabel('X')
plt.ylabel('Y')

plt.show()
```

```
[[250. 250.]
 [250. 250.]]
```



```
In [ ]:
```

Program 14

Create a GUI based form for admission purpose for your college.

```
In [1]: from tkinter import *

root = Tk()
root.title("College Admission Form")
root.geometry("400x400")

Label(root, text="Admission Form", font=("Arial", 16)).pack()

Label(root, text="Name").pack()
name = Entry(root)
name.pack()

Label(root, text="Gender").pack()
gender = StringVar()
Radiobutton(root, text="Male", variable=gender, value="Male").pack()
Radiobutton(root, text="Female", variable=gender, value="Female").pack()

Label(root, text="Course").pack()
course = Entry(root)
course.pack()

Label(root, text="City").pack()
city = Entry(root)
city.pack()

Label(root, text="Age").pack()
age = Entry(root)
age.pack()

def submit():
    print("Name:", name.get())
    print("Gender:", gender.get())
    print("Course:", course.get())
    print("City:", city.get())
    print("Age:", age.get())

Button(root, text="Submit", command=submit).pack()

root.mainloop()
```

 College Admission Form

—

□

×

Admission Form

Name

Gender

☒ Male

☐ Female

Course

City

Age

Submit

In []:

Program 15

The created GUI based application form is to connected to a database and use insert query to enter data.

```
In [1]: from tkinter import *
import sqlite3

conn = sqlite3.connect("college.db")
cursor = conn.cursor()

cursor.execute("""
CREATE TABLE IF NOT EXISTS admission (
    name TEXT,
    gender TEXT,
    course TEXT,
    city TEXT,
    age INTEGER
)
""")

root = Tk()
root.title("College Admission Form")
root.geometry("400x400")

Label(root, text="Admission Form", font=("Arial", 16)).pack()

Label(root, text="Name").pack()
name = Entry(root)
name.pack()

Label(root, text="Gender").pack()
gender = StringVar()
Radiobutton(root, text="Male", variable=gender, value="Male").pack()
Radiobutton(root, text="Female", variable=gender, value="Female").pack()

Label(root, text="Course").pack()
course = Entry(root)
course.pack()

Label(root, text="City").pack()
city = Entry(root)
city.pack()

Label(root, text="Age").pack()
age = Entry(root)
age.pack()

def submit():
    cursor.execute(
        "INSERT INTO admission VALUES (?, ?, ?, ?, ?)",
        (name.get(), gender.get(), course.get(), city.get(), age.get())
    )
    conn.commit()

Button(root, text="Submit", command=submit).pack()

root.mainloop()

conn.close()
```

College Admission Form

Admission Form

Name

Gender

☒ Male

☐ Female

Course

City

Age

Submit

In []: